Submission Summary

Conference Name

6th OPJU International Technology Conference (OTCON 6.0) on SMARTICON (Smart & Intelligent Computing for Next-Generation Technologies)

Track Name

Smart Computing for Data Science, Big Data & Analytics

Paper ID

330

Paper Title

EATALY: A Smart Cafeteria Management System with Intelligent Load Balancing and Real-Time Crowd Analytics Using Flutter and Firebase

Abstract

The increasing demand for efficient food service in educational institutions and workplaces has highlighted challenges such as long queues, overcrowding, uneven counter utilization, and inefficient order management. This paper presents EATALY, an intelligent smart cafeteria management system that integrates Flutter, Firebase, and data-driven analytics to enhance operational efficiency and user experience. The system incorporates Smart Load Balancing to dynamically distribute orders based on queue length, waiting time, and counter availability, while a Real-Time Crowd Heatmap visualizes congestion levels to assist users in selecting less crowded service areas. Additional features include Digital Token Tracking, Pickup Slot Booking, Bilingual Staff Management, Secure Authentication, Real-Time Order Synchronization, Push Notifications, Order History, and an Administrative Dashboard for monitoring cafeteria operations. Experimental evaluation using simulated scenarios demonstrated reliable cloud synchronization, effective queue distribution, and improved crowd management. The proposed architecture is scalable, cost-effective, and suitable for deployment in educational institutions, corporate campuses, hospitals, and other high-demand food service environments. Future enhancements include AI-based demand forecasting, personalized recommendations, IoT-enabled occupancy sensing, and predictive analytics to further optimize cafeteria operations.

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Submission Files

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